

Sedimentary rocks

When rocks are broken down or dissolved by weathering, the fragments are carried away by erosion and laid down as layers of sediment such as sand or mud. Over time, these layers are compressed beneath more layers of sediment, and then cemented together to form sedimentary rocks. Other sedimentary rocks, including coal and many limestones, are made from the remains of living things such as plants or marine organisms.



SANDSTONE

One of the most recognizable types of sedimentary rock, sandstone is made of grains of sand that have been cemented together, or lithified. Some sandstones form on seabeds. Others, such as this example in Utah, are lithified sand dunes made of “frosted” windblown sand laid down in cross-bedded patterns.

MUDSTONE AND SHALE

Fine sediments may be compressed and cemented together to form fine-grained mudstone. As they age, these rocks harden into shale. Mudstone and shale often contain the fossils of extinct plants and animals.

They may also contain the remains of microbes that, over millions of years, have turned into oil and natural gas.

◀ JURASSIC COAST

These mudstone cliffs on the shores of southern England are rich in fossils from the age of the dinosaurs.

FACT!

Mud carried into the Bay of Bengal by the Ganges River has built up layers of sediment and sedimentary rock that are at least 9.3 miles (15 km) deep!